

Project Plan — Smart Scientific Calculator

1. Project Overview

Field	Detail
Project Title	Smart Scientific Calculator
Development Model	Incremental Software Development Model
Technology Stack	HTML, CSS, JavaScript (Frontend) · Node.js, Express (Backend) · MongoDB via Mongoose (Database)
Team Size	[Fill in: e.g. 4 members]
Start Date	[Fill in: e.g. August 2026]
Mid-Sem Deadline	3 October 2026
Final Deadline	[Fill in]

2. Development Iterations & Timeline

The project follows six incremental iterations. Each iteration adds a well-defined set of features on top of the previous increment.

Iteration	Feature Area	Planned Start	Planned End	Status
1	Basic Arithmetic (UN-01): +, -, *, /, %, AC, DEL	Week 1	Week 2	☐ Completed
2 & 3	Scientific Functions (UN-02 & UN-03): √, sin, cos, x^y	Week 3	Week 4	☐ Completed
4	Error Handling (UN-04): Division by zero, invalid domain, syntax errors	Week 5	Week 5	☐ Completed
5	UI & Interaction (UN-05): Responsive grid layout, real-time display, button feedback	Week 6	Week 6	☐ Completed
6	History & Database (UN-06): Express server, MongoDB CRUD, history view	Week 7	Week 8	☐ Completed
—	Testing, Documentation & Final Polish	Week 9	Week 12	☐ In Progress

3. Milestone Tracker

Milestone	Target Date	Status
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Milestone	Target Date	Status
Problem statement finalised	Week 1	☐ Done
Requirements (SRS) document ready	Week 2	☐ Done
Architecture & DB design approved	Week 3	☐ Done
Iteration 1 demo-ready	End of Week 2	☐ Done
Iteration 2 & 3 demo-ready	End of Week 4	☐ Done
Iteration 4 demo-ready	End of Week 5	☐ Done
Iteration 5 demo-ready	End of Week 6	☐ Done
Iteration 6 demo-ready	End of Week 8	☐ Done
Mid-semester submission	3 Oct 2026	☐ In Progress
Final testing & polish	Week 10	☐ Planned
Final submission	[Fill in]	☐ Planned

4. Task Allocation Table

Task	Assigned To	Status
Requirements gathering & SRS	[Member 1], [Member 2]	☐ Done
Use-case diagram / User stories	[Member 2]	☐ Done
Architecture & database design	[Member 3]	☐ Done
UI wireframe & prototype	[Member 4]	☐ Done
Iteration 1 implementation	[Member 1], [Member 3]	☐ Done
Iteration 2 & 3 implementation	[Member 2], [Member 4]	☐ Done
Iteration 4 implementation	[Member 1]	☐ Done
Iteration 5 implementation	[Member 4]	☐ Done
Iteration 6 implementation	[Member 3]	☐ Done
Test-case design & execution	[Member 2], [Member 4]	☐ Done
Mid-semester report writing	All members	☐ In Progress
Demonstration video	[Member 1], [Member 3]	☐ Planned

Note: Replace [Member N] with actual team-member names.

5. Risks and Mitigation

#	Risk	Likelihood	Impact	Mitigation Strategy
1	MongoDB unavailable on deployment machine	Medium	High	Provide Docker Compose file; document manual setup steps
2	<code>eval()</code> in front-end poses security risk	Medium	Medium	Planned: replace with a safe expression parser in final phase
3	Floating-point precision in scientific functions	Low	Low	Document known tolerance; round display to 10 decimal places

#	Risk	Likelihood	Impact	Mitigation Strategy
4	Team-member unavailability near deadline	Low	High	Maintain shared repo; any member can pick up pending work
5	Moodle upload size limit exceeded	Low	Medium	Host demo video externally; include <code>DEMO_VIDEO_LINK.txt</code>

6. Delayed or Pending Tasks

Task	Original Target	Revised Target	Reason for Delay
Expression parser to replace <code>eval()</code>	Week 8	Post mid-sem	Prioritised DB integration
Keyboard input support	Week 7	Post mid-sem	Scope deferred
Unit tests with Jest (automated)	Week 9	Post mid-sem	Manual testing done first

7. Work Planned for Remaining Semester

1. Replace `eval()` with a secure mathematical expression parser.
2. Add keyboard input support for desktop users.
3. Implement automated unit tests using **Jest**.
4. Add a proper in-page history panel (instead of browser `alert()`).
5. Deploy the application to a cloud platform (e.g. Render / Railway).
6. Conduct usability testing with peer users.
7. Final documentation and submission packaging.

